Mathematics for Machine Learning

Exam Study Questions (Chapters 1–7)

Data. Run your scripts in a folder containing the data files. Lecture files: `house_prices.pkl`, `wine.csv`. Downloaded files (links in the questions and in the *Data sources* section): `anscombe.csv`, `geyser.csv`, `co2-mm-mlo.csv`, `mpg.csv`, `penguins.csv`. CSV files can be read with `np.genfromtxt(f, delimiter=",", names=True, dtype=None, encoding="utf-8")`.

Question 1

Background

Time value of money. Receiving \$1 today is worth more than receiving \$1 in a year, because money received today can be invested. If money earns interest at rate r per year (e.g. $r = 0.05$ for 5%), then \$1 today grows to $(1+r)^t$ dollars after t years. Conversely, \$1 received in year t is worth only $(1+r)^{-t}$ dollars today; this factor is called the *discount factor*.

Cash flows and NPV. A project is described by a cash-flow vector $c \in \mathbf{R}^{T+1}$, where c_t is the money received in year $t = 0, 1, \ldots, T$ (negative entries are payments, e.g. an initial investment). Collecting the discount factors in the *discount vector* $d_r = \big(1, (1+r)^{-1}, (1+r)^{-2}, \ldots, (1+r)^{-T}\big)$, the *net present value* is the value of all cash flows expressed in today's money:

$$\mathrm{NPV}_r(c) = \sum_{t=0}^{T} \frac{c_t}{(1+r)^t} = d_r^T c.$$

A project with NPV > 0 is worth more than it costs at interest rate r.

Internal rate of return (IRR). For a typical project (payments first, income later), $\mathrm{NPV}_r(c)$ decreases as r increases. The IRR is the rate r at which $\mathrm{NPV}_r(c) = 0$: the interest rate the project effectively earns.

Bisection. To solve $g(r) = 0$ for a continuous g on $[\ell, u]$ with $g(\ell)$ and $g(u)$ of opposite sign: take the midpoint $m = (\ell+u)/2$; if $g(\ell)g(m) \le 0$ the root lies in $[\ell, m]$, so set $u = m$, otherwise set $\ell = m$. Repeat until $u - \ell$ is below a tolerance (e.g. 10^{-12}). Each step halves the interval.

Two projects over $T = 5$ years (amounts in \$):

$$A = (-1000,\ 300,\ 300,\ 300,\ 300,\ 300), \qquad B = (-1000,\ 0,\ 0,\ 0,\ 0,\ 1650).$$

(a) Implement `npv(r, c)` as a single inner product with the discount vector and tabulate $\mathrm{NPV}_r(A)$, $\mathrm{NPV}_r(B)$ for $r \in \{0,\ 3\%,\ 5\%,\ 8\%,\ 12\%\}$.

(b) For fixed $r = 5\%$ the NPV is a linear function of c. Use superposition to obtain the NPV of $2A - B$ from your table and compare with evaluating $\mathrm{NPV}(2A - B)$ directly. Then give a numerical example showing that $r \mapsto \mathrm{NPV}_r(A)$ is not affine.

(c) Implement bisection and compute the IRR of A and of B (search in $[0, 1]$), and the rate $r^\star \in [0, 0.2]$ at which both projects have the same NPV. Which project is better for $r < r^\star$ and for $r > r^\star$?

Question 2

Five documents are described by word counts over the dictionary (*data, model, goal, match, team, error, coach, learning*); row i of C is the count vector c_i:

$$C = \begin{bmatrix} 6 & 4 & 0 & 1 & 0 & 3 & 0 & 2 \\ 3 & 2 & 1 & 0 & 0 & 1 & 0 & 1 \\ 0 & 1 & 5 & 4 & 3 & 0 & 2 & 0 \\ 1 & 0 & 2 & 2 & 1 & 0 & 1 & 0 \\ 4 & 3 & 1 & 1 & 0 & 2 & 1 & 1 \end{bmatrix}.$$

(a) Form the word-histogram matrix H (each row of C divided by its sum).

(b) Compute the 5×5 matrices of pairwise distances between the rows of C, between the rows of H, and of pairwise angles (in degrees) between the rows of C.

(c) Find the nearest neighbour of document 5 under (i) count distance, (ii) histogram distance, (iii) angle. Explain why the answers differ.

(d) Compute the angle matrix of H and compare it with that of C. Explain the result.

(e) Using the flop counts from the lecture, estimate the number of flops and the time (minutes on a 1 Gflop/s computer) to compute all pairwise distances for $N = 10^4$ documents and a dictionary of $n = 10^4$ words.

Question 3

The sag s (mm) at a point of a bridge is modelled as a linear function $s = c^T w$ of the loads $w \in \mathbf{R}^4$ (tonnes) at four locations. Six loading experiments gave:

exp.	w_1	w_2	w_3	w_4	measured sag s
1	1.0	0.5	0.0	0.2	0.311
2	0.0	1.2	0.4	0.0	0.476
3	0.3	0.0	1.0	0.6	0.404
4	0.5	0.5	0.5	0.5	0.435
5	2.0	0.0	1.0	0.0	0.500
6	0.4	1.0	0.0	1.5	0.628

(a) Compute the compliance vector c from experiments 1–4 (four equations, four unknowns; use `np.linalg.solve`).

(b) Predict the sags of experiments 5 and 6 and compare with the measurements.

(c) Without using c: find coefficients α with $w = (2.35, 1.5, 0, 0.6) = \sum_{k=1}^{4} \alpha_k w^{(k)}$, and predict the sag of w by superposition as $\sum_k \alpha_k s_k$. Compare with $c^T w$.

(d) Fit the affine model $s = c^T w + b$ (a possible sensor offset b) to all six experiments by least squares. Check the rank of $[\mathbf{1}\ W]$ first. Is there evidence of an offset?

Question 4

Background

Log-sum-exp and softmax. The function $f(x) = \log\left(e^{x_1} + \cdots + e^{x_n}\right)$ is a smooth approximation of $\max_i x_i$; one can show $\max_i x_i \le f(x) \le \max_i x_i + \log n$. It appears in multi-class classification and in neural networks. Its partial derivatives are

$$\frac{\partial f}{\partial x_i}(x) = \frac{e^{x_i}}{e^{x_1} + \cdots + e^{x_n}},$$

so the gradient $\nabla f(x)$ is a vector of positive numbers that sum to one, called the *softmax* of x.

Checking a gradient with finite differences. A derivative formula can be checked numerically. With the unit vector e_i and a small step h (e.g. 10^{-5}), the *central difference*

$$\frac{\partial f}{\partial x_i}(z) \approx \frac{f(z + he_i) - f(z - he_i)}{2h}$$

has an error proportional to h^2. Doing this for $i = 1, \ldots, n$ gives a numerical gradient.
Estimating an order of accuracy. If an error behaves like $E(t) \approx Ct^p$ for small t, then from two values $p \approx \log\big(E(t_2)/E(t_1)\big)/\log(t_2/t_1)$.

Let $n = 3$ and $z = (1, 2, 0)$.

(a) Implement f, its gradient (softmax), and the central-difference gradient. Compute $f(z)$ and $\nabla f(z)$ and compare with the numerical gradient.

(b) Using the lecture's first-order Taylor approximation $\hat{f}$ at z, tabulate $f(x)$, $\hat{f}(x)$ and the error $|f(x) - \hat{f}(x)|$ for $x \in \{(1.1, 2, 0), (1, 1.5, 0.5), (0, 0, 0), (3, 4, 2), (2, 1, 0)\}$.

(c) Along $x = z + t(1, -1, 0)$ compute the error for $t \in \{0.01, 0.1, 0.5, 1\}$ and estimate the order p from the first two values.

(d) Evaluate $f(x+\alpha\mathbf{1}) - f(x)$ for a random x and several α. Use the result, together with $\mathbf{1}^T \nabla f(z)$, to explain the error found at $x = (3, 4, 2)$.

Question 5

Load `house_prices.pkl` (a dict; `price` in \$1000, `area` in 1000 ft^2, `beds`, ...). The lecture uses the regression model $\hat{y} = x^T\beta + v$ with $x = (\text{area}, \text{beds})$, $\beta = (148.73, -18.85)$, $v = 54.40$.

(a) Compute all predictions with one matrix–vector product, the RMS prediction error, and the RMS error of the constant prediction $\hat{y} = \mathbf{avg}(y)$.

(b) Predict the price of an 1800 ft^2 house with 3 and with 4 bedrooms. Compute the correlation coefficient of area and bedrooms and use it to explain the negative bedroom coefficient.

(c) Compute the least squares fit with $A = [\mathbf{1}\ \text{area}\ \text{beds}]$ and compare it with the lecture parameters. Compute $A^T r$ and $\mathbf{avg}(r)$ for the least squares residual r, and explain the value of $\mathbf{avg}(r)$.

(d) Which house has the largest absolute prediction error (1-based index, price, prediction, features)?

Question 6

Background

k-nearest-neighbour regression. In the lecture, k-NN predicts a *class* by a majority vote among the k closest training points. The same idea predicts a *number*: to predict the outcome for a feature vector x, find the k data points $x^{(i)}$ closest to x (Euclidean distance) and predict the *average* of their outcomes,

$$\hat{y}(x) = \frac{1}{k} \sum_{i \in N_k(x)} y^{(i)}.$$

Leave-one-out evaluation. To judge such a predictor on a single data set without a separate test set, predict each data point i from the *other* points only, i.e. exclude i from its own neighbour set. (If point i were included, 1-NN would simply return $y^{(i)}$ and the error would be zero.) In a distance matrix this is done by setting the diagonal to $+\infty$. The RMS of the resulting prediction errors estimates the error on new data. Distance ties are broken by taking the point with the smaller index (`np.argsort(D, axis=1, kind="stable")`).

Use `house_prices.pkl`.

(a) Compute **avg**, **rms** and **std** of the price vector and check numerically that $\mathbf{rms}(x)^2 = \mathbf{avg}(x)^2 + \mathbf{std}(x)^2$.

(b) Standardize the prices. How many houses have $|z| > 3$? Give the largest and smallest z-score with the corresponding prices, and explain the asymmetry.

(c) Standardize `area`, `beds` and `baths` and stack them into feature vectors. Compute the 774×774 distance matrix and find the nearest neighbour (other than itself) of house 1.

(d) Implement leave-one-out k-NN regression of the price for $k \in \{1, 5, 10, 25, 100\}$, report the RMS prediction errors, choose k, and compare with the regression model of Question 5.

Question 7

Download `https://raw.githubusercontent.com/mwaskom/seaborn-data/master/anscombe.csv` (Anscombe's quartet: columns `dataset` $\in$ {I,II,III,IV}, `x`, `y`; background: `https://en.wikipedia.org/wiki/Anscombe%27s_quartet`).

(a) For each of the four sets compute **avg** and **std** of x and y, and the correlation coefficient.

(b) Fit a straight line $\hat{y} = \theta_1 + \theta_2 x$ to each set by least squares; report θ and the RMS fitting error. Compare θ_2 with $\rho\,\mathbf{std}(y)/\mathbf{std}(x)$.

(c) Plot the four sets with their fitted lines and describe how they differ.

(d) Set III: refit without the point with the largest absolute residual. Set IV: compute $\mathbf{std}(x)$ without the point $x = 19$. What do you conclude about summary statistics?

Question 8

Background

Portfolios. Suppose we split our money between two funds: a fraction θ in fund 1 and $1-\theta$ in fund 2. If the funds' returns in month t are a_t and b_t, the portfolio return is $p_t = \theta a_t + (1-\theta) b_t$, so the portfolio return vector is the linear combination $p(\theta) = \theta a + (1-\theta) b$. (A value $\theta < 0$ or $\theta > 1$ means borrowing one fund to buy more of the other, called *short selling*.)

Risk of a portfolio. Since **avg** is linear, $\mathbf{avg}(p) = \theta\mathbf{avg}(a) + (1-\theta)\mathbf{avg}(b)$. The risk is not linear. Writing ρ for the correlation coefficient of a and b,

$$\mathbf{std}(p)^2 = \theta^2\mathbf{std}(a)^2 + (1-\theta)^2\mathbf{std}(b)^2 + 2\theta(1-\theta)\,\rho\,\mathbf{std}(a)\mathbf{std}(b).$$

Setting the derivative with respect to θ to zero gives the *minimum-risk portfolio*

$$\theta^\star = \frac{\tilde{b}^T\tilde{b} - \tilde{a}^T\tilde{b}}{\|\tilde{a} - \tilde{b}\|^2}, \qquad \tilde{a} = a - \mathbf{avg}(a)\mathbf{1},\ \ \tilde{b} = b - \mathbf{avg}(b)\mathbf{1}.$$

If $\rho < 0$, the funds tend to move in opposite directions and their fluctuations partly cancel: this is *diversification*.

Monthly returns (%) of three (fictional) funds:

month	1	2	3	4	5	6	7	8	9	10	11	12
fund 1 (a)	1.2	−0.5	2.3	0.8	−1.4	1.9	0.4	−0.9	2.6	0.3	1.1	−0.2
fund 2 (b)	0.8	0.9	0.7	0.8	1.0	0.6	0.9	0.8	0.7	0.9	0.8	0.9
fund 3	2.1	−3.2	4.0	1.5	−2.6	3.8	0.2	−1.1	5.1	−0.4	2.7	−1.8

(a) Compute the return and risk of each fund, draw a risk–return plot, and compute the 3×3 correlation matrix.

(b) For $p(\theta) = \theta a + (1-\theta) b$ tabulate $\mathbf{avg}(p)$ and $\mathbf{std}(p)$ for $\theta \in \{0, 0.25, 0.5, 0.75, 1\}$.

(c) Implement a grid search: form all portfolios for 20001 values of θ in $[-0.5, 1.5]$ at once (one `np.outer` per fund), compute their risks and find the minimum-risk θ. Compare with the formula for $\theta^\star$ and report the return and risk of that portfolio.

(d) Check the risk formula numerically at $\theta = 0.3$, and compare $\mathbf{std}(a+b)$ with $\mathbf{std}(a) + \mathbf{std}(b)$.

Question 9

Background

From latitude/longitude to vectors. A point on the Earth is given by its latitude φ (degrees north of the equator, negative for south) and longitude λ (degrees east of Greenwich, negative for west). Model the Earth as a sphere of radius $R = 6371\,\text{km}$ centred at the origin, with the z-axis through the North Pole and the x-axis through latitude 0, longitude 0. The point is then the 3-vector

$$p = R\,\big(\cos\varphi\cos\lambda,\ \cos\varphi\sin\lambda,\ \sin\varphi\big),$$

where angles must be converted to radians (`np.radians`). Every such p has $\|p\| = R$.
Tunnel vs. surface distance. The Euclidean distance $\|p_i - p_j\|$ is the length of a straight tunnel (a chord) through the Earth. For two points at angle $\theta = \angle(p_i, p_j)$ the chord length is $2R\sin(\theta/2)$, while the spherical (great-circle) distance of the lecture is $R\theta$. For small θ, $\sin u \approx u - u^3/6$ gives $R\theta - 2R\sin(\theta/2) \approx R\theta^3/24$.

Coordinates (degrees, from Wikipedia): Istanbul $(41.01361, 28.95500)$, Ankara $(39.92889, 32.85472)$, Berlin $(52.52000, 13.40500)$, Tokyo $(35.68972, 139.69222)$, New York $(40.7128, -74.0061)$.

(a) Implement the conversion for all five cities at once and check that $\|p\| = R$.

(b) Compute the angles, spherical distances and tunnel distances for all 10 pairs.

(c) Verify the chord formula $2R\sin(\theta/2)$ numerically, and compare the Istanbul–Ankara difference (spherical − tunnel) with $R\theta^3/24$.

(d) Find Tokyo's nearest neighbour under both distances. Why must the answers agree?

Question 10

Consider the eight points in $\mathbf{R}^2$

$$x_1 = (1,1),\ x_2 = (1.5,2),\ x_3 = (3,4),\ x_4 = (5,7),$$
$$x_5 = (3.5,5),\ x_6 = (4.5,5),\ x_7 = (3.5,4.5),\ x_8 = (1,0.5).$$

Use the lecture's k-means algorithm, but with prescribed initial representatives instead of random ones.

(a) Run k-means with $k = 2$ from $z_1 = x_2$, $z_2 = x_4$. For each iteration report the assignment, the new representatives, and J^{clust} after the assignment step and after the update step. Confirm that J^{clust} never increases.

(b) Run k-means with $k = 3$ from (x_1, x_2, x_3) and from (x_1, x_3, x_4). Give the final clusters and J^{clust}.

(c) Run k-means from every possible choice of k distinct data points as initial representatives (`itertools.combinations`), for $k = 2$ and $k = 3$, and list the distinct final values of J^{clust}. What does this tell you about how k-means should be used?

Question 11

Download `https://raw.githubusercontent.com/mwaskom/seaborn-data/master/geyser.csv` (272 eruptions of the Old Faithful geyser: `duration` and `waiting` time in minutes, and a label `kind` $\in$ {long, short}). Cluster using the two numerical columns only.

(a) Compute the mean and standard deviation of both features and standardize them.

(b) Run k-means with $k = 2$ on the standardized data, starting from the first two data points. Report J^{clust}, the cluster sizes and the representatives in minutes. Cross-tabulate the clusters against `kind`.

(c) For $k = 1, \ldots, 6$ keep the best J^{clust} of 20 runs from random data points (`np.random.default_rng(0)`). Plot J^{clust} against k and choose k. Compare J^{clust} for $k = 1$ with the sum of the variances of the standardized features.

(d) Repeat (b) on the raw data. Compute the variance of each raw feature and explain which feature dominates the distances.

Question 12

Use `wine.csv` (178 wines, 13 chemical features; UCI: `https://archive.ics.uci.edu/dataset/109/wine`) and standardize all columns. Pretend that `Alcohol` (column 1) is missing for the wines in rows $1, 11, 21, \ldots, 171$ (18 wines, 1-based); the other 160 wines are complete. Use the lecture's method for guessing missing entries with k-means.

(a) Run k-means with $k = 3$ on the 160 complete wines, starting from rows 2, 72, 142. Report J^{clust} and the cluster sizes.

(b) Guess the missing `Alcohol` values and print them next to the true values (in % vol).

(c) Compare the RMS error of your guesses with that of guessing the mean alcohol of the complete wines. What happens for $k = 1$?

(d) Repeat with the best (lowest J^{clust}) of 20 random initializations (`np.random.default_rng(1)`). Report J^{clust} and the RMS error, and comment on using J^{clust} to choose the clustering for this task.

Question 13

Let $a_1 = (1, 2, 0, -1)$, $a_2 = (0, 1, 1, 2)$, $a_3 = (2, -1, 3, 1)$, $a_4 = (4, 0, 0, -7)$, $a_5 = (1, 0, 0, 1)$ and $b = (3, 1, 4, 1)$.

(a) Apply the Gram–Schmidt algorithm to a_1, a_2, a_3, a_4, report $\|\tilde{q}_i\|$ at every step, and decide whether the vectors are linearly independent.

(b) Find coefficients γ with $a_4 = \gamma_1 a_1 + \gamma_2 a_2 + \gamma_3 a_3$ (least squares) and check the residual.

(c) Apply Gram–Schmidt to a_1, a_2, a_3, a_5. Are they a basis of $\mathbf{R}^4$? Compute the expansion of b in this basis and multiply the coefficients by 22 to recognize exact fractions.

(d) Compute the rank of $[a_1\ a_2\ a_3\ a_4]$ and check that $a_1 + a_2 + a_3 + a_4 = 3a_1 - 2a_2 + 2a_3$. Why does this not contradict the uniqueness of coefficients from the lecture?

Question 14

Background

Functions as vectors. Sample a function g on the grid $t_1, \ldots, t_{201}$ of 201 equally spaced points in $[-1, 1]$ (`t = np.linspace(-1, 1, 201)`); the samples form a vector in $\mathbf{R}^{201}$. The inner product of two sampled functions, $\sum_i g(t_i)h(t_i)$, is (up to the factor 0.01) a Riemann sum for $\int_{-1}^{1} g(t)h(t)\,dt$. Functions with zero inner product are called *orthogonal*.

Legendre polynomials. Applying Gram–Schmidt to the functions $1, t, t^2, t^3$ with the integral inner product gives, after scaling each to the value 1 at $t = 1$, the Legendre polynomials $P_0 = 1$, $P_1 = t$, $P_2 = \frac{1}{2}(3t^2 - 1)$, $P_3 = \frac{1}{2}(5t^3 - 3t)$. With the sampled (discrete) inner product one gets slightly different polynomials. To read off the coefficients of a sampled cubic q, fit a cubic to it with

`np.polynomial.polynomial.polyfit(t, q, 3)` (coefficients ordered $1, t, t^2, t^3$).
Best approximation by orthonormal vectors. If $q_1, \dots, q_m$ are orthonormal, the vector in their span closest to b is $\hat{b} = \sum_k \alpha_k q_k$ with $\alpha_k = q_k^T b$. In matrix form, with the q_k as rows of Q, $\alpha = Qb$ and $\hat{b} = Q^T\alpha$. The residual $r = b - \hat{b}$ is orthogonal to every q_k ($Qr = 0$), and Pythagoras gives $\|b\|^2 = \|\alpha\|^2 + \|r\|^2$.

(a) Orthonormalize the sampled vectors $1, t, t^2, t^3$ with Gram–Schmidt into the rows of $Q \in \mathbf{R}^{4\times 201}$ and check $\max |QQ^T - I|$.

(b) Scale each q_k to the value 1 at $t = 1$, find its polynomial coefficients and compare with $P_0, \dots, P_3$.

(c) For $b = |t|$ and $b = e^t$, compute α, the residual norm, RMS and maximum absolute error, $\max |Qr|$, and $\|b\|^2$ vs. $\|\alpha\|^2 + \|r\|^2$. Explain why two of the α_k vanish for $|t|$, and why e^t is approximated much better.

Question 15

Background

Vandermonde determinant. For the square Vandermonde matrix V with rows $(1, t_i, t_i^2, \dots, t_i^{n-1})$, $i = 1, \dots, n$, one has

$$\det V = \prod_{1 \le i < j \le n} (t_j - t_i),$$

which is nonzero exactly when all t_i are distinct. In that case there is exactly one polynomial of degree $\le n-1$ through any n points (t_i, y_i).
Calculus as matrix multiplication. Store $p(t) = c_1 + c_2 t + \cdots + c_n t^{n-1}$ as $c \in \mathbf{R}^n$. Its derivative $p'(t) = c_2 + 2c_3 t + \cdots + (n-1)c_n t^{n-2}$ has coefficient vector Dc, where D is the $(n-1) \times n$ matrix with entries $D_{i,i+1} = i$ and zeros elsewhere; for $n = 4$:

$$D = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 3 \end{bmatrix}.$$

Since $\int_0^1 t^{k-1} dt = 1/k$, the integral is an inner product: $\int_0^1 p(t)\, dt = w^T c$ with $w = (1, \frac{1}{2}, \dots, \frac{1}{n})$. (Hint: `np.diag(v, k=1)` places v on the first superdiagonal.)

(a) Build the Vandermonde matrix for $t = (-1, 0, 1, 2)$ and find the cubic through $(-1, 2.5)$, $(0, 1)$, $(1, 0.5)$, $(2, 7)$.

(b) Compare $\det V$ with the product formula, and compute the rank of V when $t = (-1, 0, 0, 2)$.

(c) Evaluate the cubic at $t = (-2, -0.5, 0.5, 1.5, 3)$ with a single matrix–vector product.

(d) Implement D and w for general n. Compute Dc and $\int_0^1 p$ for your cubic, and the number of nonzeros and the density of D for $n = 4$ and $n = 100$.

Question 16

Use the advertising example of the lecture, with reach matrix (1000 views per \$; rows = 10 demographic groups, columns = 3 channels)

$$R^T = \begin{bmatrix} 0.97 & 1.23 & 0.80 & 1.29 & 1.10 & 0.67 & 0.87 & 1.10 & 1.92 & 1.29 \\ 1.86 & 2.18 & 1.24 & 0.98 & 1.23 & 0.34 & 0.26 & 0.16 & 0.22 & 0.12 \\ 0.41 & 0.53 & 0.62 & 0.51 & 0.69 & 0.54 & 0.62 & 0.48 & 0.71 & 0.62 \end{bmatrix}.$$

(a) Reproduce the lecture's least squares spending for $v^{\text{des}} = 10^3\mathbf{1}$ and its RMS error.

(b) A new campaign targets 2000 in groups 1–5 and 500 in groups 6–10. Compute $\hat{s}$, the achieved views $R\hat{s}$, the RMS error and the relative RMS error $\mathbf{rms}(R\hat{s} - v^{\text{des}})/\mathbf{rms}(v^{\text{des}})$. Also compute $\hat{s} = (R^TR)^{-1}R^Tv^{\text{des}}$ by solving $(R^TR)s = R^Tv^{\text{des}}$ and compare.

(c) Compute R^Tr for the optimal residual $r = R\hat{s} - v^{\text{des}}$ and interpret it geometrically.

(d) Since $\hat{s} = R^{\dagger}v^{\text{des}}$ with a fixed matrix $R^{\dagger}$, the solution depends linearly on the target. Compute $\hat{s}$ for the targets $(\mathbf{1}_5, 0)$ and $(0, \mathbf{1}_5)$ and combine them to reproduce (b).

(e) The solution in (b) has a negative entry. Refit using only channels 2 and 3 and compare the RMS error.

Question 17

Monthly mean CO_2 concentration (ppm) at Mauna Loa (NOAA GML, `https://gml.noaa.gov/ccgg/trends/data.html`; file `https://raw.githubusercontent.com/datasets/co2-ppm/main/data/co2-mm-mlo.csv`). The header has fewer names than columns; read it with `np.genfromtxt(f, delimiter=",", skip_header=1, usecols=(1, 2))` (decimal date, monthly mean) and keep January 2015 – December 2025:

year	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
2015	400.18	400.55	401.74	403.35	404.15	402.97	401.46	399.11	397.82	398.49	400.27	402.06
2016	402.73	404.25	405.06	407.60	407.90	406.99	404.59	402.45	401.23	401.79	403.72	404.64
2017	406.36	406.66	407.54	409.22	409.89	409.08	407.33	405.32	403.57	403.82	405.31	407.00
2018	408.15	408.52	409.59	410.45	411.44	410.99	408.90	407.16	405.71	406.19	408.21	409.27
2019	411.03	411.96	412.18	413.54	414.86	414.15	411.96	410.17	408.76	408.74	410.47	411.97
2020	413.59	414.32	414.72	416.42	417.28	416.58	414.58	412.75	411.50	411.49	413.10	414.23
2021	415.49	416.72	417.61	419.01	419.09	418.93	416.90	414.42	413.26	413.90	414.97	416.67
2022	418.13	419.24	418.76	420.19	420.97	420.94	418.85	417.15	415.91	415.74	417.47	418.99
2023	419.48	420.31	420.99	423.35	424.00	423.68	421.83	419.68	418.50	418.82	420.46	421.86
2024	422.80	424.55	425.38	426.51	426.90	426.91	425.55	422.99	422.03	422.38	423.85	425.40
2025	426.65	427.09	428.15	429.64	430.51	429.61	427.87	425.48	424.37	424.87	426.46	427.49

Index the months $i = 1, \ldots, 132$ (January 2015 = 1). Train on $i \le 120$ (2015–2024), test on 2025.

Background

Quadratic trend. The lecture's model $\hat{y} = \hat{y}^{\text{lin}} + \hat{y}^{\text{seas}}$ has a constant growth rate θ_1 per month. To allow the growth rate to change over time, add the basis function i^2: $\hat{y}_i = \theta_1 i + \theta_2 i^2 +$ (seasonal value of month i). The growth rate at month i is then $\theta_1 + 2\theta_2 i$, so $\theta_2 > 0$ means accelerating growth.

De-seasonalized trend. Removing the seasonal oscillation from a seasonal model gives the long-term level $\theta_1 i$ $(+\theta_2 i^2) + \bar{s}$, where $\bar{s}$ is the mean of the 12 seasonal values. Extrapolating this curve tells us when the long-term level reaches a given value.

(a) Fit the straight-line trend $\hat{y}_i = \theta_1 + \theta_2 i$. Report the growth in ppm/year and the training and test RMS errors.

(b) Fit the lecture's trend + seasonal model with $P = 12$. Report the growth in ppm/year, the seasonal pattern minus its mean (months of maximum and minimum, peak-to-peak), and the RMS errors. Compute the rank of the matrix when a constant column is added, and explain.

(c) Fit the quadratic trend + seasonal model. Report θ_1, θ_2, the RMS errors, and the growth rate at $i = 120$.

(d) Print the 2025 predictions of both seasonal models next to the measurements and comment.

(e) Using the de-seasonalized trend, find the month in which 450 ppm is reached under each seasonal model.

Question 18

Download `https://raw.githubusercontent.com/mwaskom/seaborn-data/master/mpg.csv` (UCI Auto MPG, `https://archive.ics.uci.edu/dataset/9/auto+mpg`). Drop the 6 cars with missing `horsepower`, keeping the original order ($N = 392$). Let $y =$ `mpg` and $x =$ `horsepower`$/100$. For reproducible results, use 5-fold cross-validation with fixed folds instead of a random shuffle: car i (0-based) belongs to fold $i \bmod 5$. Report the RMS cross-validation error $\sqrt{(\epsilon_1^2 + \cdots + \epsilon_5^2)/5}$.

(a) Fit polynomials of degree $0, 1, \ldots, 6$. Tabulate the RMS training error (fit on all data) and the RMS cross-validation error.

(b) Choose a degree following the lecture's advice, give the fitted model, and predict the mpg of a 130 hp car.

(c) Fit the piecewise-linear model with knots at 100 and 150 hp. Report the slope of each of the three pieces (mpg per 100 hp) and both RMS errors.

(d) Add the feature `weight`$/1000$ to the degree-2 model. Report θ and both RMS errors, and interpret the weight coefficient.

Question 19

Download `https://raw.githubusercontent.com/mwaskom/seaborn-data/master/penguins.csv` (Palmer penguins, `https://allisonhorst.github.io/palmerpenguins/`). Features: bill length, bill depth, flipper length (mm), body mass (**kg**); drop the 2 penguins with missing values ($N = 342$). Test set: penguins with 0-based index $i \bmod 5 = 0$ (69 penguins); training set: the other 273. Label $y = +1$ for Chinstrap, $y = -1$ otherwise.

Background

Changing the decision threshold. The least squares classifier predicts $\operatorname{sign}(\tilde{f}(x))$, i.e. it compares $\tilde{f}(x)$ with the threshold 0. More generally we can use

$$\hat{f}_\alpha(x) = \operatorname{sign}\big(\tilde{f}(x) - \alpha\big).$$

Lowering α makes the classifier answer $+1$ more often, which increases the true positive rate but also the false positive rate; raising α does the opposite. Plotting (FPR, TPR) for many values of α gives the *receiver operating characteristic* (ROC) curve. The threshold is chosen according to how costly each type of error is in the application.

A baseline. The *trivial classifier* $\hat{f}(x) = -1$ ignores the features. Its error rate equals the fraction of positive examples; a useful classifier must do clearly better.

(a) Fit the Boolean least squares classifier on the training set and report θ.

(b) Give the confusion matrices on the training and test sets, with error rate, true positive rate, false positive rate and precision.

(c) Implement the threshold classifier $\hat{f}_\alpha$ and evaluate it on the test set for $\alpha \in \{-0.5, -0.25, 0, 0.25, 0.5\}$. Which α would you use if missing a Chinstrap is costly?

(d) Refit with bill length as the only feature and compare its test performance with the trivial classifier.

(e) Classify a new penguin with $x = (49.0, 18.5, 195, 3.7)$ and comment on the confidence.

Question 20

Use the same data, features and training/test split as in Question 19, now with the three classes Adelie (0), Chinstrap (1) and Gentoo (2).

Background

Distance concentration. The lecture explains that in high dimensions the nearest and the farthest points are almost equally far away. This can be measured with a simulation: draw 1000 points uniformly in the cube $[0,1]^d$ and one more *query* point (`rng.random((1000, d))`, `rng.random(d)`), compute the 1000 distances from the query point, and form the ratio

$$\frac{\text{smallest distance}}{\text{largest distance}} \in (0,1].$$

A ratio close to 0 means the nearest neighbour is much closer than the rest; a ratio close to 1 means all points are about equally far away, so the nearest neighbour carries little information.

(a) Fit the lecture's multi-class least squares classifier (one-hot targets, prediction by arg max) on the training set. Give the 3×3 test confusion matrix and the numbers of training and test errors.

(b) Classify the test set with the lecture's k-NN classifier for $k = 1, 5, 15$ using (i) the raw features (mass in kg), (ii) features standardized with the *training* means and standard deviations, (iii) the raw features with mass in grams. Report the numbers of test errors and explain the differences.

(c) What does k-NN predict for $k = 273$ (all training points), and what is its test error rate?

(d) Compute $\ell = 0.01^{1/d}$ for $d = 1, 2, 4, 10, 50, 100$ and the smallest d with $\ell \geq 0.9$. Then implement the distance-concentration simulation (`np.random.default_rng(0)`) for $d = 2, 10, 100, 1000$, and relate the results to k-NN.

Data sources

- Questions 5, 6: `house_prices.pkl`; Question 16: reach matrix — lecture notes / lecture data.
- Question 7: `https://raw.githubusercontent.com/mwaskom/seaborn-data/master/anscombe.csv`; `https://en.wikipedia.org/wiki/Anscombe%27s_quartet`
- Question 9 coordinates (accessed 20 Oct 2026): `https://en.wikipedia.org/wiki/Istanbul`, `https://en.wikipedia.org/wiki/Ankara`, `https://en.wikipedia.org/wiki/Berlin`, `https://en.wikipedia.org/wiki/Tokyo`, `https://en.wikipedia.org/wiki/New_York_City`
- Question 11: `https://raw.githubusercontent.com/mwaskom/seaborn-data/master/geyser.csv`
- Question 12: `wine.csv`; `https://archive.ics.uci.edu/dataset/109/wine`
- Question 17: `https://gml.noaa.gov/ccgg/trends/data.html`; `https://raw.githubusercontent.com/datasets/co2-ppm/main/data/co2-mm-mlo.csv` (accessed 20 Oct 2026)
- Question 18: `https://raw.githubusercontent.com/mwaskom/seaborn-data/master/mpg.csv`; `https://archive.ics.uci.edu/dataset/9/auto+mpg`
- Questions 19, 20: `https://raw.githubusercontent.com/mwaskom/seaborn-data/master/penguins.csv`; `https://allisonhorst.github.io/palmerpenguins/`
- Questions 2, 3, 8: small data sets made up for these exercises.